AI Mobile Hackathon 2026

Team Details

Team Name
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Problem Statement
TrustCircle: AI-Powered Relationship Health App

Presentation Outline

- Introduction
- Literature Review
- Problem Definition
- Objectives
- Methodology
- Proposed Outcomes
- Conclusion
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Introduction: TrustCircle

Overview of the Topic

TrustCircle is an AI-powered mobile application designed to monitor and improve relational health within close-knit groups. By leveraging on-device AI for daily anonymous emotional check-ins, it identifies relational patterns and provides proactive nudges to strengthen interpersonal trust before conflict arises.

Close relationships are the single most significant determinant of long-term mental health. Current solutions target individual wellness or reactive therapy, leaving a critical gap in *preventive relational health*.

Importance & Key Findings

Global Mental Health
1 in 8 people live with a mental health condition. (WHO, 2023)

Social Isolation
43% of adults feel lonely and disconnected from close relationships. (APA, 2022)

Economic Impact
Workplace disengagement costs $8.8 trillion annually due to broken trust. (Gallup, 2023)

Indian Household Stress
Joint/nuclear family stress has escalated post-pandemic with no tailored digital tools.

Literature Review

Topic	Paper	Year	Summary	Key Gap
NLP & Relationship Quality Inference	K. Loveys et al., "Microblog language use and relationship quality in couples: A computational analysis," *IEEE Transactions on Affective Computing*, vol. 13, no. 4, pp. 2041–2053.	2022	• Demonstrated that short, low-friction text inputs are sufficient for meaningful relational quality inference via NLP. • Validated that invasive behavioral surveillance is not required for accurate relational health detection. • Achieved statistically significant correlation between language patterns and self-reported relationship quality.	• Limited to romantic dyads using public social media text; not private or group-applicable. • No on-device or privacy-preserving architecture. • No real-time intervention or conversation starter generation.
AI for Stress & Wellbeing Detection	M. K. Hasan, A. Choudhury, and R. Picard, "Detecting stress and wellbeing using passive sensing and AI: A systematic review," *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 6, pp. 2793–2807.	2023	• Systematic review showing hybrid models (self-report + AI inference) outperform either method alone. • Found passive sensing combined with AI achieves clinically meaningful wellbeing detection rates. • Validated multi-modal input design as the strongest approach for health inference.	• All reviewed systems target individuals, not relational groups. • No anonymized group-level aggregation model proposed. • Privacy-preserving architecture absent from most reviewed systems.

Literature Review

Topic	Paper	Year	Summary	Key Gap
LLMs in Mental Health Support	E. C. Stade et al., "Large language models could change the future of behavioral health," *NEJM AI*, vol. 1, no. 4.	2024	• Demonstrated that well-prompted LLMs generate therapeutically appropriate supportive responses comparable to trained counselors on certain tasks. • Validated AI-generated conversation starters as a legitimate supplementary intervention mechanism. • Identified LLMs as scalable tools for preventive mental health support.	• Studies conducted in individual therapy contexts; no group relational health application. • Hallucination risk and cultural sensitivity in non-Western contexts remain open problems.
Workplace Disengagement & Trust	Gallup, *State of the Global Workplace: 2023 Report*. Gallup Press.	2023	• Quantified that workplace disengagement costs $8.8 trillion annually, largely attributable to broken interpersonal trust. • Found only 23% of employees report feeling engaged at work globally. • Established economic case for proactive team trust monitoring as a business priority.	• Measures organizational sentiment at a macro level; no tool for small-team interpersonal trust circles. • No anonymous, privacy-safe mechanism for peer-level trust monitoring proposed.

Literature Review

Topic	Paper	Year	Summary	Key Gap
Relational Health & Longevity	R. J. Waldinger and M. S. Schulz, *The Good Life: Lessons from the World's Longest Scientific Study of Happiness*. Simon & Schuster.	2023	• 80-year Harvard longitudinal study establishing close relationships as the strongest predictor of long-term health and happiness. • Quantified that social connection outperforms genetics, wealth, and career as a health determinant. • Validated relational quality as a clinical priority, not merely a social concern.	• Study is observational and retrospective; no real-time monitoring or early intervention mechanism proposed. • No mobile or AI-assisted tool for proactive relational maintenance suggested.
Digital Mental Health in India	S. Banerjee and P. Sharma, "Digital mental health interventions in India: Barriers, facilitators and future directions," *Indian Journal of Psychiatry*, vol. 65, no. 3, pp. 312–320.	2023	• Documented escalation of family and interpersonal stress in Indian households post-pandemic. • Found existing digital tools are Western-designed with poor cultural fit — no collectivist framing, no family-circle model. • Identified a significant gap in Hindi-language and culturally grounded relational health tools.	• No existing tool addresses joint/nuclear Indian family dynamics. • No AI-powered group-level relational health monitoring solution proposed for Indian context.

Key Understandings

The Technological Convergence ("How")

Synthesis: Short daily emotional inputs + NLP + Gemini Nano are sufficient for relational health inference — no invasive monitoring.

The Shift: Reactive tools to proactive, AI-assisted daily rituals that detect trust erosion early.

The Privacy-Utility Paradox ("Gap")

Prisoner's Dilemma: Shared data is needed, but exposing individual responses destroys psychological safety.

Missing Link: No system provides anonymous aggregated group-level insight while preserving individual privacy.

The "Heterogeneity Hurdle"

Insight: Dynamics vary by culture and relationship type; one-size-fits-all is insufficient.
Objective: Design culturally sensitive, context-aware AI conversation starters.

Conclusion: Shift relationship health from reactive management to proactive daily maintenance.

The Three-Fold Problem ("The Why")

- **Privacy Conflict:** Anonymity vs. Shared Insight.
- **Reactivity Trap:** Intervening after damage is done.
- **Coverage Vacuum:** No single framework for family, friends, and teams.

"Designing a system that delivers collective relational insight with the accuracy of shared data but the safety of complete individual anonymity — bridging the gap between Group Intelligence and Personal Privacy."

Problem Definition

Problem Statement

To design an AI-powered mobile app that proactively monitors collective relational health through anonymous daily check-ins and AI pattern detection, enabling preventive intervention before trust erosion becomes irreversible.

Context: The Invisible Deterioration

Relationships fail through accumulated micro-failures. Current solutions are reactive and siloed. No existing tool:

- Covers all relationship types (family, friends, work)
- Uses multi-user AI pattern detection
- Maintains individual anonymity
- Is designed for mobile-first global users

Significance of the Gaps

Clinical: Prevent damage before it leads to clinical mental health issues.

Technological: Treat the bond as the unit of care, not just the individual.

Economic: Mitigate ₹6.5 lakh crore loss due to broken team trust.

Ethical: Provide safety through absolute data privacy.

"Addressing the failure of reactive relationship management by building a framework for collective insight without compromising individual vulnerability."

Objectives

Main Objective

To design, develop, and validate an AI-powered cross-platform mobile application (Flutter) that proactively monitors and improves the relational health of close-knit groups using anonymous daily emotional check-ins, on-device and cloud AI pattern detection, and real-time voice-guided resolution support.

Specific Objectives

1. Emotional Pulse Check-in

Anonymous daily system with Riverpod & Firebase Firestore sync, ensuring zero exposure of individual scores.

2. Dual-Gemini Integration

Gemini 2.0 Flash (cloud) & Nano (on-device) for nightly pattern analysis and AI conversation starters.

3. Voice-Based Resolution

Agora ConvoAI SDK for natural language emotional input with secure on-device processing.

4. Relational Reinforcement

Longitudinal "gratitude chain" and silence detector to flag disengagement before conflict arises.

Methodology

Design Approach

This project employs a ***mixed-methods, iterative agile design*** approach combining quantitative AI model evaluation with qualitative user experience research. The development follows a 4-phase sprint structure across 7 weeks.

Data Collection Methods

Primary: Daily anonymous slider-based emotional check-ins (4 dimensions) and voice transcripts via Gemini Nano.

Secondary: User satisfaction surveys (Week 3 & 7) and semi-structured interviews with 3–5 test circles.

Passive: Engagement tracking (frequency, gratitude rate, duration) via Firebase Analytics.

Data Analysis Techniques

Quantitative: Stats on trust trends, Pearson correlations, and A/B completion rate comparisons.

AI Evaluation: Blind panel rating of Gemini 2.0 outputs and on-device inference benchmarking.

Qualitative: Thematic analysis of transcripts using open coding to identify friction and adoption patterns.

"Producing testable deliverables validated by real users in test circles through a robust 4-phase sprint structure."

Tools & Resources

App Layer

Frontend | **Flutter 3.22+ / Dart** | Cross-platform Android & iOS development

State Management | **Riverpod 2.x** | Reactive, compile-safe application state

Voice AI | **Agora ConvoAI SDK** | Real-time voice check-in, guided resolve sessions

AI & Intelligence

On-device AI | **Gemini Nano (AICore)** | On-device sentiment inference, privacy preservation

Cloud AI | **Gemini 2.0 Flash** | Pattern detection, conversation starter generation

Sentiment Analysis| **ML Kit Text API** | On-device gratitude message tone classification

Backend & Data

Database | **Firebase Firestore** | Real-time multi-user data synchronisation

Authentication | **Firebase Auth** | Secure circle invite and user management

Backend Logic | **Cloud Functions (Node.js)** | Nightly AI analysis, aggregation, FCM triggers

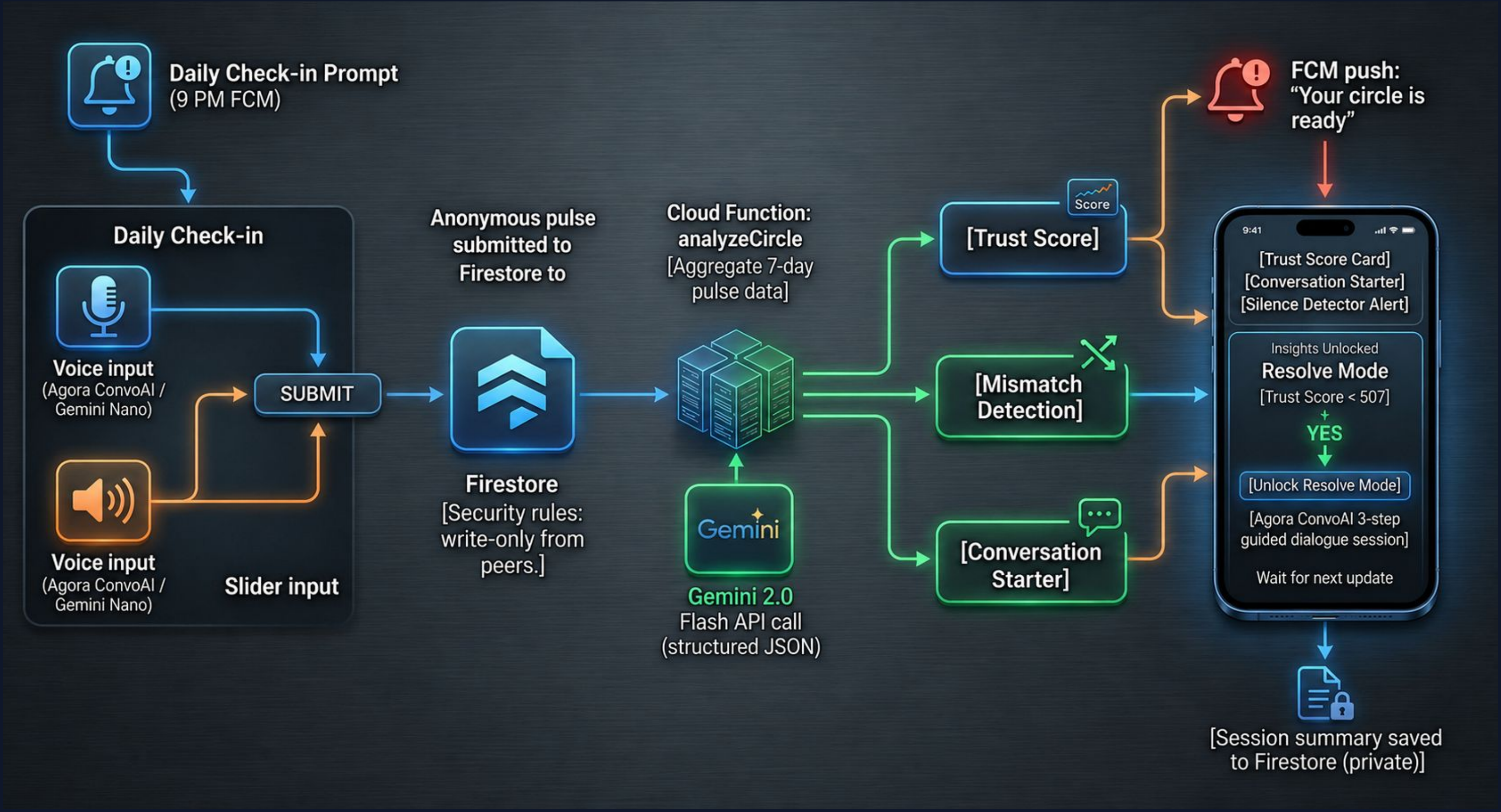
DevOps & Growth

CI/CD | **GitHub Actions** | Automated build, test, deployment pipeline

Analytics | **Firebase Analytics** | Engagement tracking, funnel analysis

Notifications | **Firebase Cloud Messaging** | Daily check-in nudges, insight-ready alerts

Methodology Flowchart



Proposed Outcomes

Expected Results

The primary output is a production-ready mobile app (Android + iOS) for groups of 2–6 to monitor relational health via AI rituals.

- **Trust Accuracy:** ≥75% correlation with satisfaction scores.
- **Voice Check-in:** ≥15% higher completion vs. slider-only.
- **Gratitude Chain:** ≥60% daily active participation.
- **Silence Detector:** ≥80% precision on disengagement alerts.

Potential Impact on the Field

Clinical: Scalable preventive layer to reduce clinical escalations.

Technical: Replicable privacy architecture for anonymous health apps.

Social: Culturally grounded design for Indian collectivist structures.

Academic: Empirical data contribution to affective computing.

Future Research Directions

- Passive behavioral signals (latency, tone) for richer inference.
- Longitudinal study (6–12 months) on clinical measures.
- Integration with wearable biosensors (HRV tracking).
- Federated learning via Gemini Nano for data privacy.

Conclusion

Summary of Key Points

TrustCircle addresses the silent erosion of relational trust using anonymous daily check-ins, on-device AI (Gemini Nano), cloud patterns (Gemini 2.0 Flash), and real-time voice (Agora ConvoAI).

The project synthesises attachment and self-determination theories with state-of-the-art AI, serving as an academic contribution, a hackathon entry, and a viable product.

Final Thoughts

The most important infrastructure of a healthy society is relational, not technological. TrustCircle proposes that AI, built with privacy and empathy, catalyzes human connection rather than substituting for it.

The application makes conversation more likely, honest, and constructive.

Call to Action / Next Steps

- Complete MVP build (Jun 4)
- Deploy to TestFlight & Google Play Internal Testing
- Submit to AI Mobile Hackathon 2026 Finale (Jun 21)
- Draft IEEE conference paper (DroidCon / FlutterCon)
- File provisional patent for AI aggregation architecture

References

[1] K. Loveys, G. Fricchione, K. Bhatt, and E. Torous, "Microblog language use and relationship quality in couples: A computational analysis," *IEEE Transactions on Affective Computing*, vol. 13, no. 4, pp. 2041–2053, Oct.–Dec. 2022, doi: 10.1109/TAFFC.2022.3161098.

[2] M. K. Hasan, A. Choudhury, and R. Picard, "Detecting stress and wellbeing using passive sensing and AI: A systematic review," *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 6, pp. 2793–2807, Jun. 2023, doi: 10.1109/JBHI.2023.3261748.

[3] E. C. Stade et al., "Large language models could change the future of behavioral health," *NEJM AI*, vol. 1, no. 4, Apr. 2024, doi: 10.1056/AIra2400802.

[4] Gallup, *State of the Global Workplace: 2023 Report*. Washington, DC, USA: Gallup Press, 2023.

[5] R. J. Waldinger and M. S. Schulz, *The Good Life: Lessons from the World's Longest Scientific Study of Happiness*. New York, NY, USA: Simon & Schuster, 2023.

[6] S. Banerjee and P. Sharma, "Digital mental health interventions in India: Barriers, facilitators and future directions," *Indian Journal of Psychiatry*, vol. 65, no. 3, pp. 312–320, Mar. 2023, doi: 10.4103/indianjpsychiatry.indianjpsychiatry_56_23.

Thank You!